YETI PAW · CABLE HANDLE · RESEARCH BOOK

TWO PIVOT POSITIONS, FOUR MOVEMENTS

The YETI PAW can be built with the cable hooked on in one of two places. The current pivot puts most of the load on the forearm pad, nearer a wrist cuff. The centered pivot shares it between hand and pad, nearer a D-ring handle. This book runs that comparison on the lateral raise, the bicep curl, the Bayesian curl and the chest fly, and ends with the hardware it led to. Both pivots go to a lifter trial as a hypothesis, not a verdict.

V2.1 · 2026-10-17 · CABLE SOURCE: VOLTRA, 45° RACK BRACKET, EXIT 8 IN. UP · FLY: FLUSH-MOUNTED PAIR JUST BELOW SHOULDER HEIGHT

THE BOTTOM LINE

13–17 lb

Current: forearm-led. Of a 20 lb cable, the forearm pad holds 13–17 lb at the hardest point of every movement. The grip can relax mid-rep only: at the start of the bicep curl, Bayesian curl, cable chest fly the current pivot's grip force goes above the 20 lb a D-handle puts in the hand (up to 24 lb).

6–8 lb

Centered: balanced. At the same moment the pad holds 6–8 lb and the hand the rest, nearer a D-ring handle.

6–14% less

The same peak at about 6–14% less weight. At equal weight the centered pivot's peak cable torque at the working joint is higher, so the same peak takes less on the Voltra.

MOVEMENT	HARDEST POINT	CURRENT · PAD + HAND	CENTERED · PAD + HAND	CENTERED PEAK, SAME WEIGHT	SAME PEAK AT
Lateral raise	44°	17 + 3 lb	**8 + 12 lb**	+7%	6% less
Bicep curl	115°	13 + 7 lb	**6 + 14 lb**	+16%	14% less
Bayesian curl	89°	13 + 7 lb	**6 + 14 lb**	+16%	14% less
Cable chest fly	31°	17 + 3 lb	**8 + 12 lb**	+7%	6% less

Reading the table: 20 lb on the cable, at the current pivot's hardest point; both pivots are compared at that same moment (each contact's share along the cable). The last two columns compare each pivot's own peak. The raise and the fly, and the two curls, come out alike because the numbers are device geometry plus limb length: the raise and fly use the same 33 in. arm, the curls the same 15 in. forearm. Peak pad and grip forces over the whole rep are on page 5.

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What changed in v2.1 (after a critical review of v2.0)

- Yoke swing reported as an envelope over every stance checked; where it passes the stop, and that the cable pulls at full tension there.
- Small / median / large lifter table: the headline numbers are device geometry plus limb length.
- Peak pad push and grip force over the whole rep (true contact forces), not only the hardest-point share; “the grip can relax” qualified.
- Pad lift-off decided from the sign of the pad's contact force; values printed uncapped; the hand is never said to lift off.
- Wrist moment at the hardest point and its peak, for all four handles.
- Stretched-position row per movement; the “same curve shape” claims dropped where the data contradict them.
- Cuff comparison over a grid of cuff-ring positions; “follows a cuff most closely” dropped where the ranking flips.
- Handle tilted ±10° and ±20° on the arm; confidence wording downgraded.
- Wording: “same peak at N% less weight”, cable torque (not “effort”), trial hypothesis with drop criteria, hardware strength and 75 lb rating.

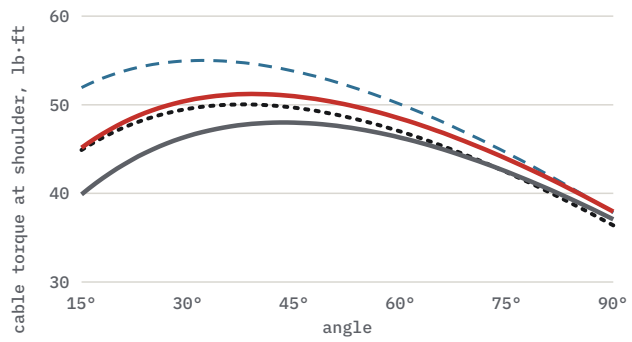
How sure are we? Confident on the direction and rough size, not on exact pounds or feel. One 2D physics model of one measured lifter (6′5″), worked twice from scratch; the stances for the curls and the fly are assumed. Lifter size and how the handle sits on the arm move the pounds (page 6); elbow bend in raises and flies is not modeled.

The colors: **red** is the centered pivot and the forearm pad; gray the current pivot; **blue** the hand and the D-handle; dotted black a wrist cuff. Every comparison is made at the current pivot's hardest point. Next step: 3–5 lifters try both variants back to back, with a wrist cuff for reference; a variant drops out if lifters can’t tell it apart, if most prefer the other, or if its pad clunks where the numbers say it may lift off.

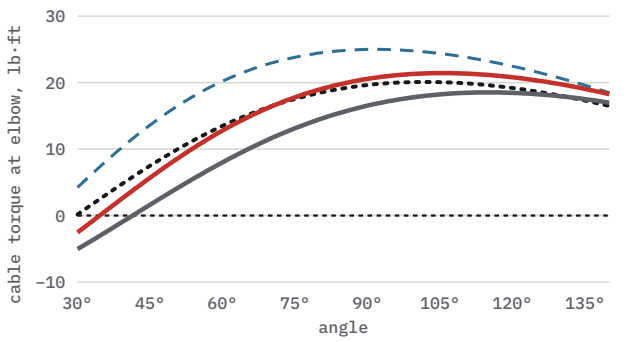
RESISTANCE CURVES

Cable torque at the working joint through each movement, for 20 lb on the cable: how hard the cable turns the limb about the joint, in lb·ft (pounds times feet of leverage). The limb's own weight is not included. Below zero the cable is helping the limb along.

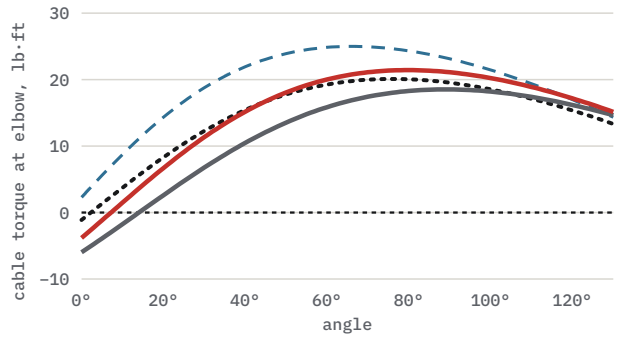
LATERAL RAISE



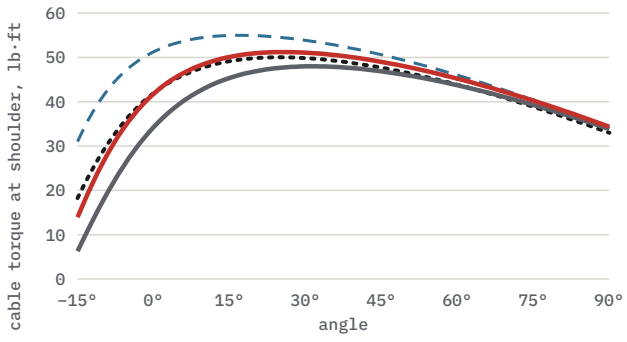
BICEP CURL



BAYESIAN CURL



CABLE CHEST FLY



Wrist cuff · D-handle · YETI PAW · current · YETI PAW · centered pivot

PEAK CABLE TORQUE, LB·FT	CURRENT	CENTERED	WRIST CUFF	D-HANDLE	CENTERED VS CURRENT
Lateral raise	48.0 at 44°	**51.2 at 39°**	50.0 at 38°	55.0 at 32°	+7%
Bicep curl	18.5 at 115°	**21.4 at 106°**	20.1 at 102°	25.0 at 92°	+16%
Bayesian curl	18.5 at 89°	**21.4 at 80°**	20.1 at 76°	25.0 at 66°	+16%
Cable chest fly	48.0 at 31°	**51.2 at 26°**	50.0 at 25°	55.0 at 17°	+7%

Reading across: the centered pivot peaks 5–9° earlier than the current one and 7–16% higher. Tables round to whole or tenth lb·ft; every percentage is taken from the unrounded values.

STRETCHED POSITION	START ANGLE	CURRENT	CENTERED	WRIST CUFF	D-HANDLE
Lateral raise	15°	39.9	**45.1**	44.9	51.9
Bicep curl	30°	−5.0	**−2.5**	0.2	4.2
Bayesian curl	0°	−6.0	**−3.8**	−1.1	2.3
Cable chest fly	−15°	6.3	**13.9**	18.3	31.0

Reading the table: cable torque in lb·ft at the first angle of each rep. The handles load the stretched position very differently, so the curves do not share one shape: on the fly the D-handle loads the start about 5× as much as the current pivot. That is a real training difference between the handles, not a flaw in any of them.

GAP TO A CUFF CURVE	CURRENT · RING 3″/1.2″	CENTERED · RING 3″/1.2″	CURRENT · RING GRID	CENTERED · RING GRID	RANKING FLIPS
Lateral raise	4.4%	**2.6%**	1.8–5.7%	**0.7–9.1%**	yes
Bicep curl	19.5%	**7.2%**	9.1–23.5%	**2.1–22.3%**	yes
Bayesian curl	18.1%	**7.5%**	9.1–21.8%	**2.2–22.2%**	yes
Cable chest fly	9.3%	**2.9%**	3.7–11.4%	**0.9–9.1%**	yes

Reading the table: RMS gap between each pivot's torque curve and a wrist cuff's, as % of the cuff's peak, with the cuff ring 2.5–5 in. behind the fist center and 0.8–2 in. below the wrist. With the ring where the studies assume it (3 in. back, 1.2 in. down) the centered pivot's curve is the closer one on every movement. Within 2.5–5 in. back and 0.8–2 in. down the ranking flips on every movement, so neither pivot can be said to follow a cuff more closely in general. Keep two meanings apart: on where the load sits, the current pivot is nearer a cuff and the centered nearer a D-handle; on the shape of the curve, the answer depends on where the cuff ring sits.

PAD, GRIP AND WRIST LOADS

Peak contact forces over the whole rep, for 20 lb on the cable. These are the full pushes on the pad and the grip, squeeze included, so they run higher than the shares on page 1 (the squeeze is felt, but it doesn't add to the cable load).

MOVEMENT	CURRENT · PEAK PAD PUSH, LB	PEAK GRIP, LB	PAD LIFTS OFF	CENTERED · PEAK PAD PUSH, LB	PEAK GRIP, LB	PAD LIFTS OFF
Lateral raise	23 at 15°	18 at 90°	never	**13 at 15°**	**20 at 90°**	**88–90°**
Bicep curl	23 at 67°	24 at 30°	never	**14 at 58°**	**22 at 30°**	**never**
Bayesian curl	23 at 41°	24 at 0°	128–130°	**14 at 32°**	**22 at 0°**	**118–130°**
Cable chest fly	23 at 0°	24 at −15°	never	**14 at −6°**	**21 at 90°**	**79–90°**

Reading the table: The grip can relax mid-rep only: at the start of the bicep curl, Bayesian curl, cable chest fly the current pivot's grip force goes above the 20 lb a D-handle puts in the hand (up to 24 lb). The pad lifts off (its push turns negative) on the lateral raise with the centered pivot at 88–90°; on the Bayesian curl with the current pivot at 128–130°; on the Bayesian curl with the centered pivot at 118–130°; on the cable chest fly with the centered pivot at 79–90°. The handle then hangs on the grip alone, and the pad landing again on the way back may be felt as a clunk. A D-handle puts the full 20 lb in the grip all rep.

WRIST MOMENT, LB·IN	CURRENT	CENTERED	WRIST CUFF	D-HANDLE
Lateral raise	9 / 33	**34 / 43**	0 / 0	59 / 60
Bicep curl	17 / 42	**38 / 42**	0 / 0	56 / 60
Bayesian curl	17 / 42	**38 / 46**	0 / 0	56 / 60
Cable chest fly	9 / 40	**35 / 45**	0 / 0	59 / 60

Reading the table: at the hardest point / peak over the rep. The bending load the hand puts on the wrist in the plane of the movement, about a wrist joint 3 in. behind the fist center along the forearm (lb·in: pounds times inches of leverage). Where the pad lifts off, the hand carries the whole handle. The cuff strap is taken to sit on the forearm side of the wrist, so it puts none on the wrist.

LIFTER SIZE AND FIT

The headline numbers are device geometry plus limb length. Limb: elbow to fist (curls) 12 / 13.5 / 15 in., or the straight arm scaled with it (raise, fly; 33 in. measured at 15 in.). Shoulder and cable exit stay put.

MOVEMENT	LIFTER · LIMB	HARDEST	CURRENT · PAD + HAND	CENTERED · PAD + HAND	CENTERED VS CURRENT, PEAK
Lateral raise	Small 26.4″	53°	16 + 4 lb	**8 + 12 lb**	+9%
	Median 29.7″	48°	16 + 4 lb	**8 + 12 lb**	+8%
	Large 33.0″	44°	17 + 3 lb	**8 + 12 lb**	+7%
Bicep curl	Small 12″	126°	11 + 9 lb	**4 + 16 lb**	+20%
	Median 13.5″	120°	12 + 8 lb	**5 + 15 lb**	+17%
	Large 15″	115°	13 + 7 lb	**6 + 14 lb**	+16%
Bayesian curl	Small 12″	98°	10 + 10 lb	**4 + 16 lb**	+20%
	Median 13.5″	93°	12 + 8 lb	**5 + 15 lb**	+17%
	Large 15″	89°	13 + 7 lb	**6 + 14 lb**	+16%
Cable chest fly	Small 26.4″	44°	16 + 4 lb	**8 + 12 lb**	+9%
	Median 29.7″	38°	16 + 4 lb	**8 + 12 lb**	+8%
	Large 33.0″	31°	17 + 3 lb	**8 + 12 lb**	+7%

Reading the table: pad + hand shares of a 20 lb cable at each lifter's hardest point. Across sizes the centered peak runs 7–20% above the current one.

HANDLE TILTED ON THE ARM (±10°, ±20°)	CURRENT · PAD SHARE	CENTERED · PAD SHARE
Lateral raise	8–21 lb	**3–12 lb**
Bicep curl	4–21 lb	**0–11 lb**
Bayesian curl	4–20 lb	**1–11 lb**
Cable chest fly	8–21 lb	**3–12 lb**

Reading the table: the pad's share of a 20 lb cable at the hardest point when the handle–pad line is turned 10° or 20° either way off the limb, as it might sit on a real forearm. It moves the pounds more than any stance guess: direction and rough size, not exact pounds. Elbow bend is not modeled.

Designed by Schnibbz · YETI STRENGTH · YETI PAW research v2.1 · 2026-10-17

YETI PAW · CABLE LATERAL RAISE · DESIGN STUDY

LATERAL RAISE: TWO WAYS TO CARRY THE LOAD

The YETI PAW can be built with the cable hooked on in one of two places. Both work, and they suit different lifters. The current pivot puts most of the load on the forearm pad. A centered pivot hooks on near the wrist and shares the load between hand and pad, closer to a D-ring handle; the current pivot, with most of it on the forearm, sits nearer a wrist cuff. Both go to a lifter trial as a hypothesis.

THE BOTTOM LINE

17 + 3 lb

Current: forearm-led. At the hardest point of the movement the forearm pad carries 17 lb of a 20 lb cable and the hand 3 lb. The grip can relax mid-rep: at the hardest point it takes 4 lb (full grip force). Its grip force peaks at 18 lb (at 90°), under the 20 lb a D-handle puts in the hand.

8 + 12 lb

Centered: balanced. The same moment: 8 lb on the pad, 12 lb in the hand. The load sits between hand and pad instead of mostly on the forearm, closer to a D-ring handle.

Plates only

Small change to make. Only the side plates change: the yoke hole moves between handle and pad. Matches the original plate in the same 2D check (static 4.4, fatigue 1.5–1.6). Rated 75 lb; the yoke is the weakest part. A Voltra can be set above 75 lb, so don't. The yoke needs −62° to +122° across all four movements (plate: −81° to +103°), so it rides on its stop early in some curls and flies.

Besides the split, the cable torque at the shoulder differs: the centered pivot's is up to 14% higher at the start and 2% at the top; the same peak comes at about 6% less weight.

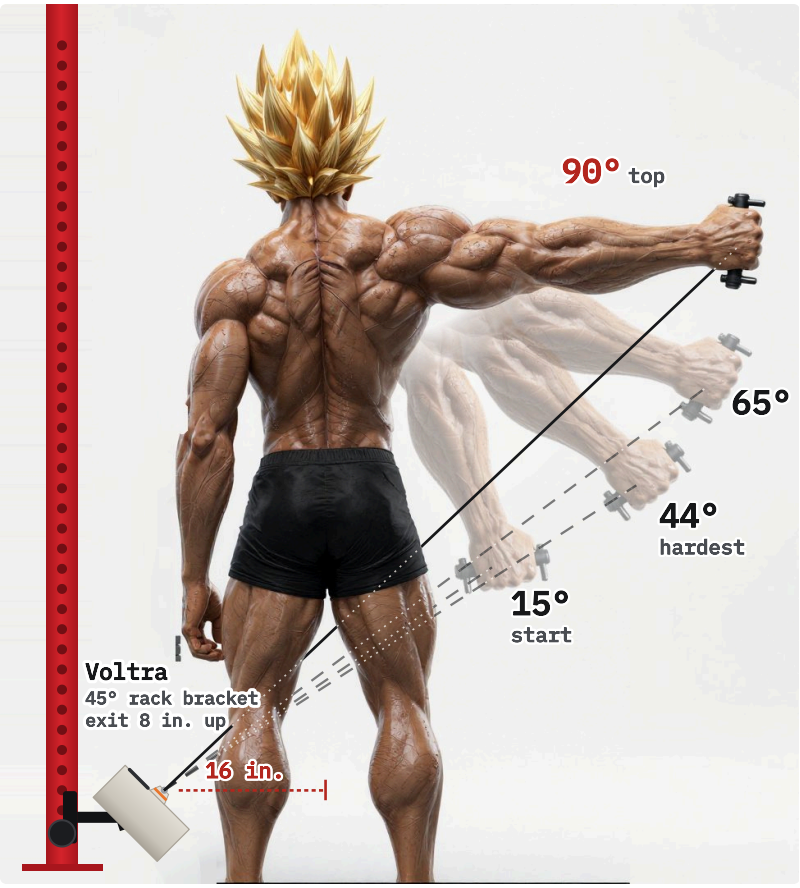
How sure are we? Confident on the direction and rough size, not on exact pounds or feel. One 2D model of one measured lifter (6′5″), worked twice (matched within 1%). Stance guesses move the pad's share at the hardest point 15–18 lb (current); tilting the handle 10–20° on the arm moves it 8–21 lb (current), 3–12 lb (centered). Elbow bend is not modeled.

Nobody has used a centered-pivot prototype yet. The centered pivot's pad lifts off at 88–90°: the handle then hangs on the grip alone, and the pad lands again on the way back, which may be felt as a clunk.

Next step: 3–5 lifters try both variants back to back, with a wrist cuff for reference. A variant drops out if lifters can't tell it apart, if most prefer the other, or if its pad clunks where the numbers say it may lift off.

THE TEST MOVEMENT

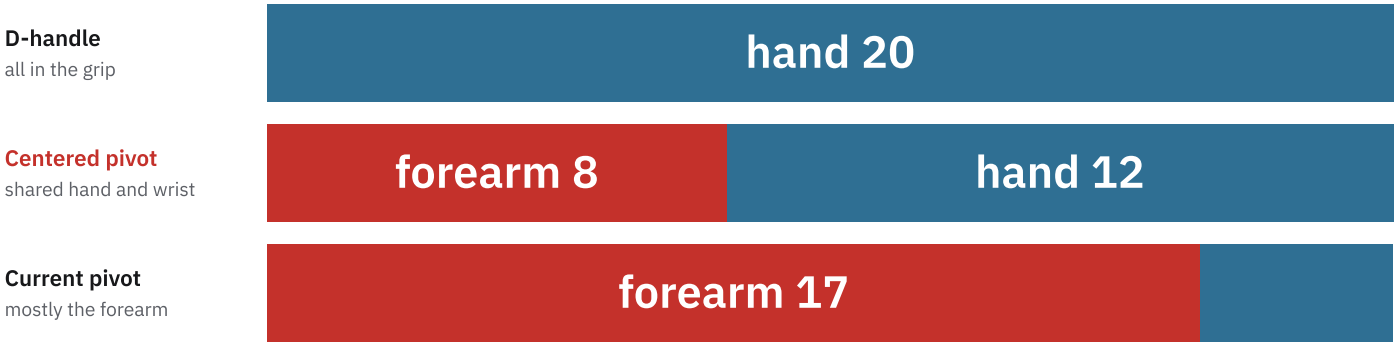
The comparison needs one movement that swings the arm through a wide range of angles against the cable, so a cable lateral raise is used as the yardstick. It is a measuring stick, not the point: the same shift between hand and forearm shows up in any movement where the cable pulls across the forearm, though the exact pounds depend on the angle. Seen from behind, the cable runs from a Voltra on a 45° rack bracket beside the left foot (cable exit 8 in. up), across the front of the body, to the right hand; the arm rises from 15° to 90°.



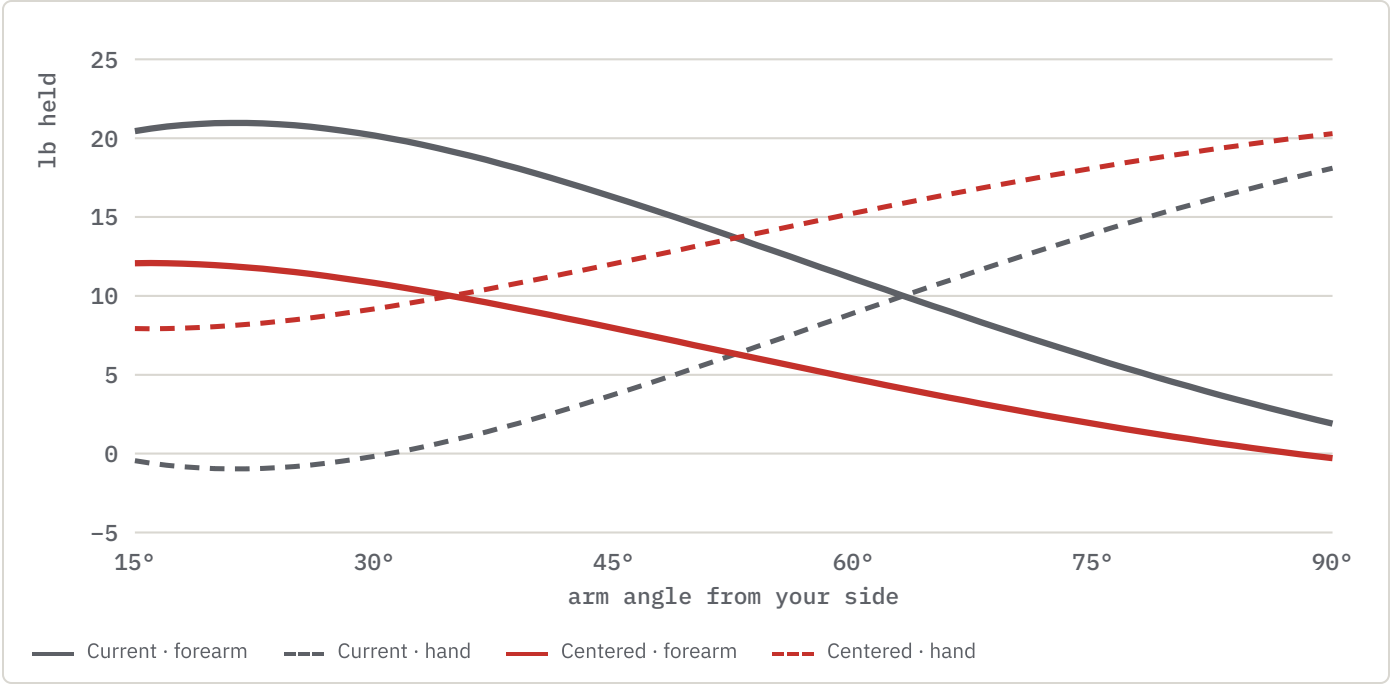
Drawn to scale from the model: true arm length, true cable-exit position, every arm position pivoting about the same shoulder. The cable passes in front of the body, so from behind it is hidden (dotted).

WHERE THE 20 LB SITS

At the hardest point of the test movement (44°), here is how a 20 lb cable is carried, with a plain D-handle for reference.



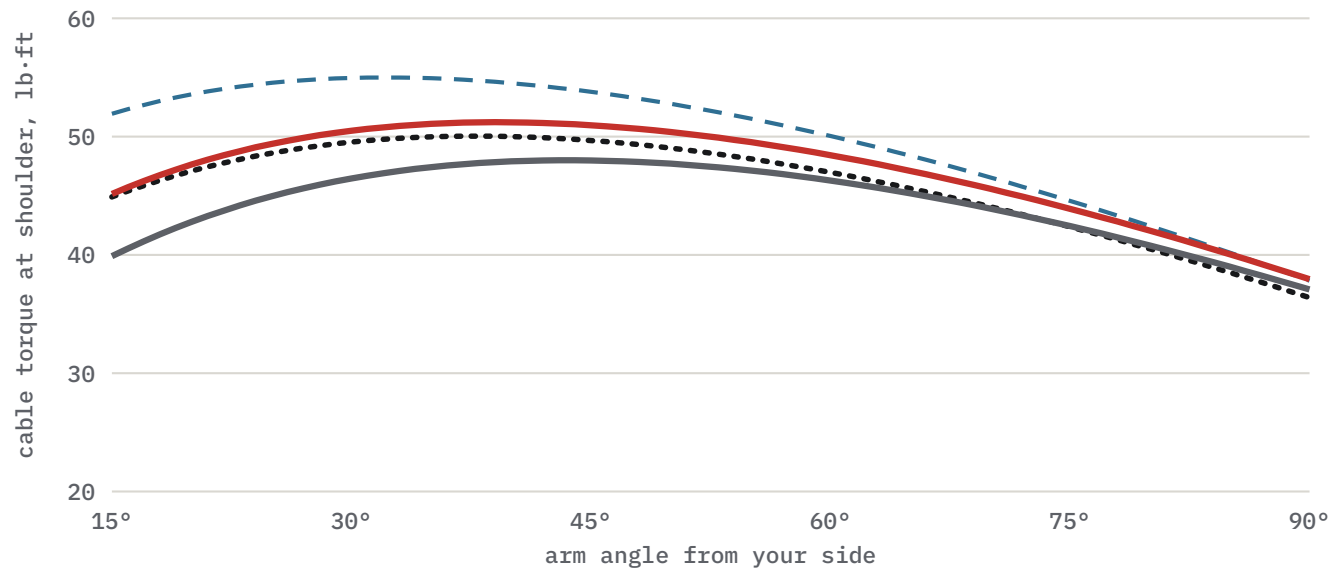
Reading the bars: red is the forearm pad's share of the cable, blue the hand's; each bar adds up to the 20 lb cable. A wrist cuff puts everything on the wrist strap. The two YETI PAW variants sit at different points between a D-handle and full forearm support.



Reading the chart: the same split through the whole movement. At every arm angle the solid line (forearm pad) and the dashed line of the same color (hand) add up to 20 lb. The current pivot (gray) is forearm-led for most of the movement and hands over to the grip near the top. The centered pivot (red) keeps hand and pad closer to even through the middle.

AGAINST A CUFF AND A D-HANDLE

Two different comparisons, kept apart. **Where the load sits:** the current pivot, with most of it on the forearm pad, is nearer a wrist cuff; the centered pivot, sharing it with the grip, is nearer a D-handle. **The shape of the resistance curve:** see below. Where the cable hooks on shapes how the resistance builds and fades through the movement. With the cuff ring where we assumed it (3 in. behind the fist center, 1.2 in. below the wrist), the centered pivot's curve is the closer one: an RMS gap of 2.6% of the cuff's peak (centered) against 4.4% (current). Move the ring anywhere from 2.5–5 in. back and 0.8–2 in. down and the ranking flips: centered 0.7–9.1%, current 1.8–5.7%. So neither pivot can be said to follow a cuff more closely in general.



At the start of the rep (15°), the stretched position, the four handles differ: cable torque at the shoulder is 39.9 lb·ft with the current pivot, 45.1 centered, 44.9 with a wrist cuff and 51.9 with a D-handle. How much the stretched position is loaded is a real training difference between the handles, not a flaw in any of them.

Reading the chart: cable torque at the shoulder: how hard the cable turns the arm about the joint, in lb·ft (pounds times feet of leverage; the arm's own weight is not included). The dotted black line is a wrist cuff with its ring 3 in. back. The red line (centered pivot) runs alongside it the whole way, a few percent above. The gray line (current pivot) starts a little lower and meets it near the top. A plain D-handle (blue dashes) is highest at the start because the cable hangs on the very end of the arm.

Both to trial

Trial hypothesis, not a verdict. Both pivots go to the lifter trial as a hypothesis: the current pivot for lifters who want the forearm to carry the load, the centered pivot for a balanced, D-handle-like feel. A variant drops out if lifters can’t tell it apart from the other, if most prefer the other in a back-to-back trial, or if its pad lifts off and clunks where the numbers say it might. Same parts, different side plates.

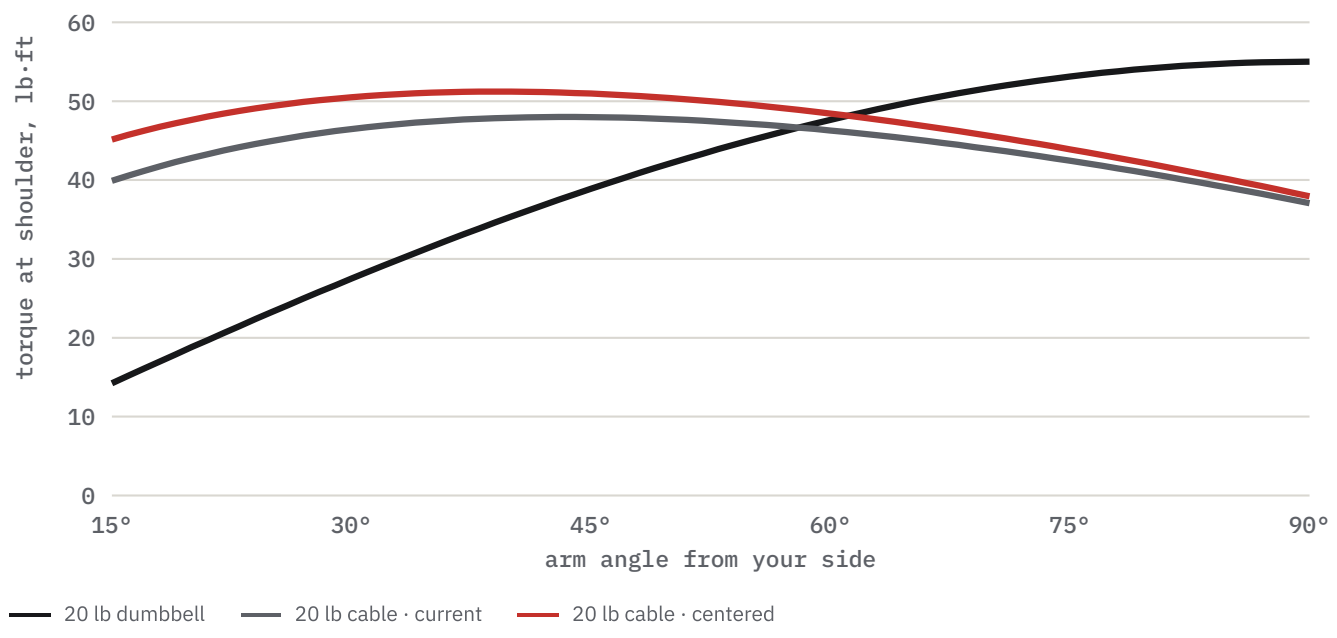
APPENDIX · CABLE TORQUE NUMBERS

ANGLE	DUMBBELL	D-HANDLE	WRIST CUFF	CENTERED	CURRENT
15°	14	52	45	45	40
30°	27	55	50	50	46
45°	39	54	50	51	48
60°	48	50	47	48	46
75°	53	45	42	44	42
90°	55	38	36	38	37

Reading the table: cable torque at the shoulder in lb·ft (pounds times feet of leverage; arm weight not included) for a 20 lb load at each arm angle of the test movement. The dumbbell column is a 20 lb dumbbell, for comparison.

APPENDIX · CABLE VERSUS DUMBBELL

For context: a cable from the floor beside the opposite foot is a different exercise from a dumbbell. A dumbbell pulls straight down, so it is easy at the bottom and hardest with the arm level. The cable pulls across the arm from the start, so it is about 3× harder at the bottom and eases off near the top. Every number in this study is for the cable.



Reading the chart: black is a 20 lb dumbbell, gray and red are a 20 lb cable on the two YETI PAW handles. Higher means more torque at the shoulder.

APPENDIX · HAND AND FOREARM, IN DETAIL

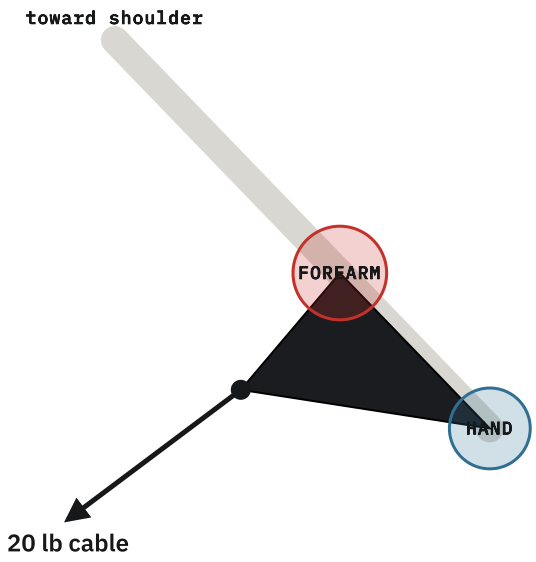
HOLDING

Hand and forearm together hold the cable. The two always add up to exactly 20 lb.

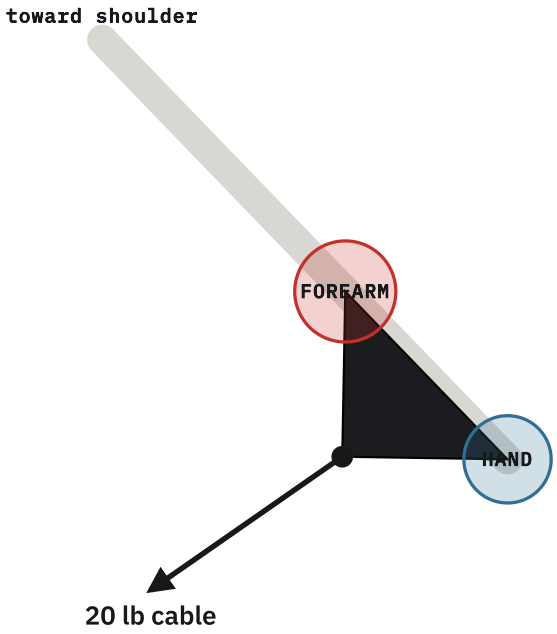
SQUEEZE

The cable hooks on to one side, so it tries to twist the handle. Hand and forearm press against each other to stop it. You feel it, but it doesn't add to the cable load.

CURRENT — PIVOT BELOW THE WRIST PAD, 44°



CENTERED — PIVOT BETWEEN HAND AND PAD, 44°



ANGLE	CURRENT · SHARE PAD / HAND	PAD PUSH	GRIP	CENTERED · SHARE PAD / HAND	PAD PUSH	GRIP
15°	20 / −0.5	23	10	12 / 8	13	10
30°	20 / −0.2	20	3	11 / 9	11	9
45°	16 / 4	16	4	8 / 12	8	12
60°	11 / 9	12	10	5 / 15	5	15
75°	6 / 14	7	15	2 / 18	2	18
90°	2 / 18	3	18	−0.3 / 20	−0.4 lifts off	20

Reading the table: pounds at each arm angle. Share = each contact's part of the 20 lb cable (the two add up to 20). Pad push and grip = the full contact forces, squeeze included. A negative pad push means the pad lifts off.

HANDLE	PEAK PAD PUSH, LB	PEAK GRIP FORCE, LB	PAD LIFTS OFF	WRIST MOMENT AT HARDEST, LB·IN	PEAK WRIST MOMENT, LB·IN
YETI PAW · current	23 at 15°	18 at 90°	never	9	33 at 90°
YETI PAW · centered	13 at 15°	20 at 90°	88–90°	34	43 at 90°
D-handle	—	20 all rep	—	59	60 at 32°
Wrist cuff	—	0 strap 20	—	0	0 at 15°

Reading the table: 20 lb on the cable, over the whole rep. Pad push and grip force are the full contact forces, squeeze included, so they can add up to more than 20 lb (the squeeze cancels between them). The pad lifts off where its push turns negative. Wrist moment: the bending load the hand puts on the wrist joint, taken 3 in. behind the fist center along the forearm, in the plane of the movement (lb·in: pounds times inches of leverage). The cuff strap is taken to sit on the forearm side of the wrist, so it puts none on the wrist.

APPENDIX · HOW THESE NUMBERS WERE MADE

MEASURED

- Lifter 6′5″
- Cable exit 8″ off the floor (a Voltra on a 45° rack bracket), about 10″ out from the outside of the left leg
- Fist center 29″ off the floor, arm hanging
- Top of the outside of the delt 64″ off the floor
- Handle holes 120 / 140 / 85 mm (checked against a photo)

ASSUMED

- Shoulder joint center 2″ below the delt top, 8″ out from the midline
- Outside of the leg 6″ from the midline at cable-exit height, so the cable exit sits 16″ left of center
- Straight arm, 33″ shoulder to fist
- Wrist cuff ring 3″ behind the fist center, 1.2″ below the wrist line; wrist joint 3″ behind the fist center
- Handle–pad line along the arm; no elbow bend
- Raise in the plane of the body, seen from behind
- Pad presses square to the forearm; the grip does the rest

CHECKS

- Cable torque = cable force × its distance from the shoulder joint; the cable pulls from the yoke pivot toward the cable exit
- Re-derived independently from scratch; every number matched within 1%
- Moving the cuff ring anywhere from 2.5–5″ back and 0.8–2″ down: RMS gap to the cuff curve 0.7–9.1% (centered) vs 1.8–5.7% (current); which is closer flips
- Handle tilted ±10° and ±20° on the arm: pad share at the hardest point 8–21 lb (current) vs 3–12 lb (centered)
- Moving the shoulder or cable-exit guesses by several inches: the forearm share at the hardest point stays 15–18 lb (current) vs 7–9 lb (centered)

Designed by Schnibbz · YETI STRENGTH · YETI PAW research v2.1 · 2026-10-17

YETI PAW · CABLE CURL · FACING THE VOLTRA · DESIGN STUDY

BICEP CURL: TWO WAYS TO CARRY THE LOAD

The same comparison as the lateral raise, on the bicep curl. The current pivot puts most of the load on the forearm pad. The centered pivot hooks on near the wrist and shares the load between hand and pad, closer to a D-ring handle. Both go to a lifter trial as a hypothesis.

THE BOTTOM LINE

13 + 7 lb

Current: forearm-led. At the hardest point of the bicep curl (115°) the forearm pad holds 13 lb of a 20 lb cable and the hand 7 lb. Over the rep the pad pushes up to 23 lb and the grip up to 24 lb.

6 + 14 lb

Centered: balanced. The same moment: 6 lb on the pad, 14 lb in the hand. Over the rep the pad pushes up to 14 lb and the grip up to 22 lb. A plain D-handle puts all 20 in the hand.

−14% weight

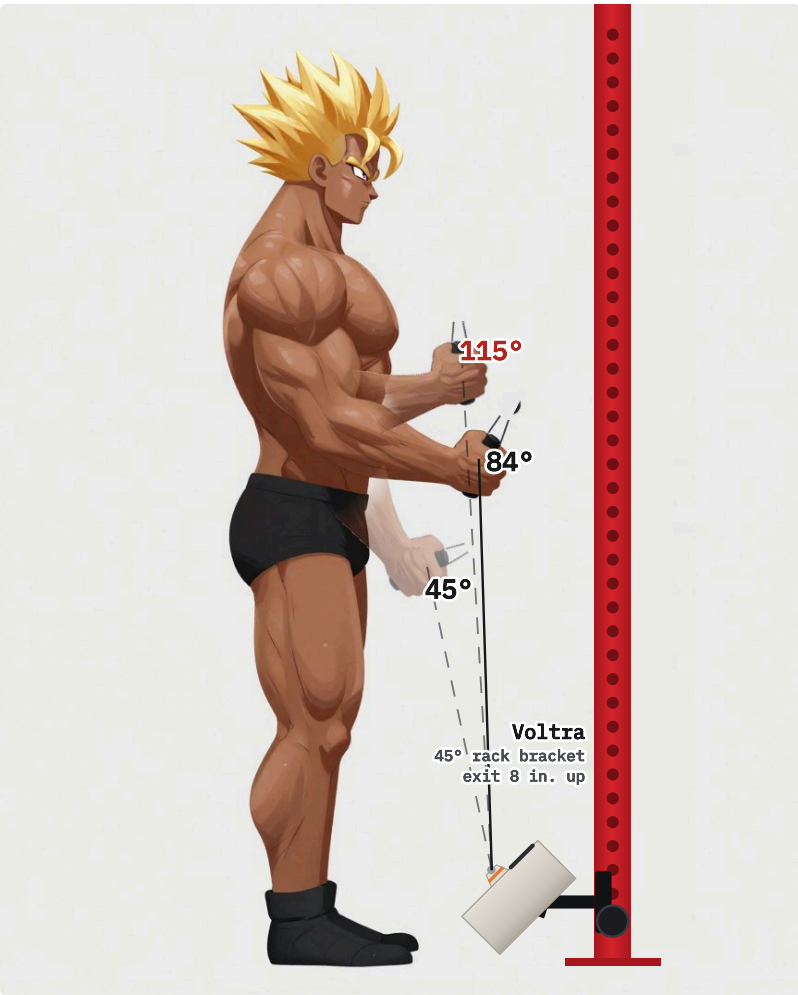
The same peak at about 14% less weight. At equal weight the centered pivot's peak cable torque at the elbow is 16% higher than the current pivot's (a wrist cuff: +8%), so the same peak takes about 14% less on the Voltra.

How sure are we? Confident on the direction and rough size, not on exact pounds or feel. The numbers come from the same 2D physics model as the lateral raise study, built on one real lifter (6′5″, measured), with the stance and forearm length assumed (see the appendix). Across the stances checked, the forearm holds 12–14 lb at the hardest point with the current pivot and 5–6 lb with the centered one. Tilting the handle 10–20° on the arm moves it more: 4–21 lb (current), 0–11 lb (centered).

The grip can relax mid-rep: at the hardest point it takes 8 lb (full grip force). Not at the ends: from 30–52° the current pivot's grip takes up to 24 lb, more than the 20 lb a D-handle puts in the hand. Wear the handle so the pad sits on the side of the forearm away from the cable, as on the lateral raise.

THE MOVEMENT

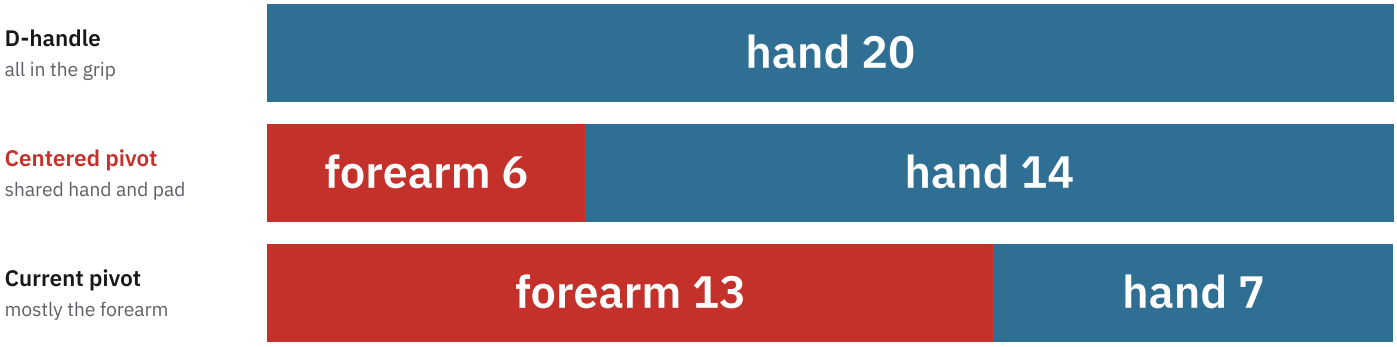
A standing cable curl, seen from the side, facing a Voltra on a 45° rack bracket: its cable exits 8 in. off the floor, about 16 in. in front of the elbow. The elbow stays at the side while the forearm rises. Early in the curl the cable runs nearly along the forearm and helps rather than resists: the YETI PAW starts carrying load at 42° (current pivot) or 35° (centered). The sheet follows the curl from 30° to 140°, measured from the arm hanging straight.



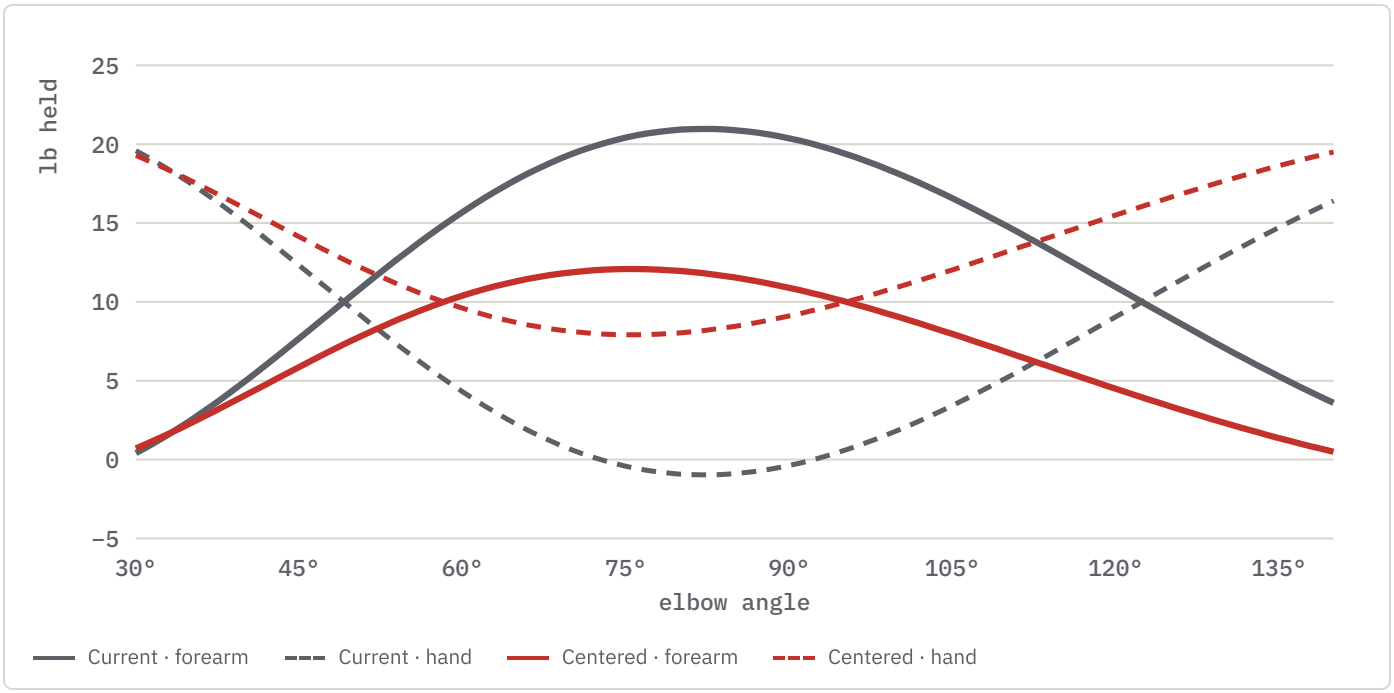
Drawn to the model: one forearm, one length, pivoting about one elbow at the exact angles shown (solid = the pose as drawn, faded = other points in the curl), and the Voltra's cable exit where the model puts it, 8 in. off the floor he stands on. The red angle is the hardest point.

WHERE THE 20 LB SITS

At the hardest point (115°), here is how a 20 lb cable is carried, with a plain D-handle for reference.



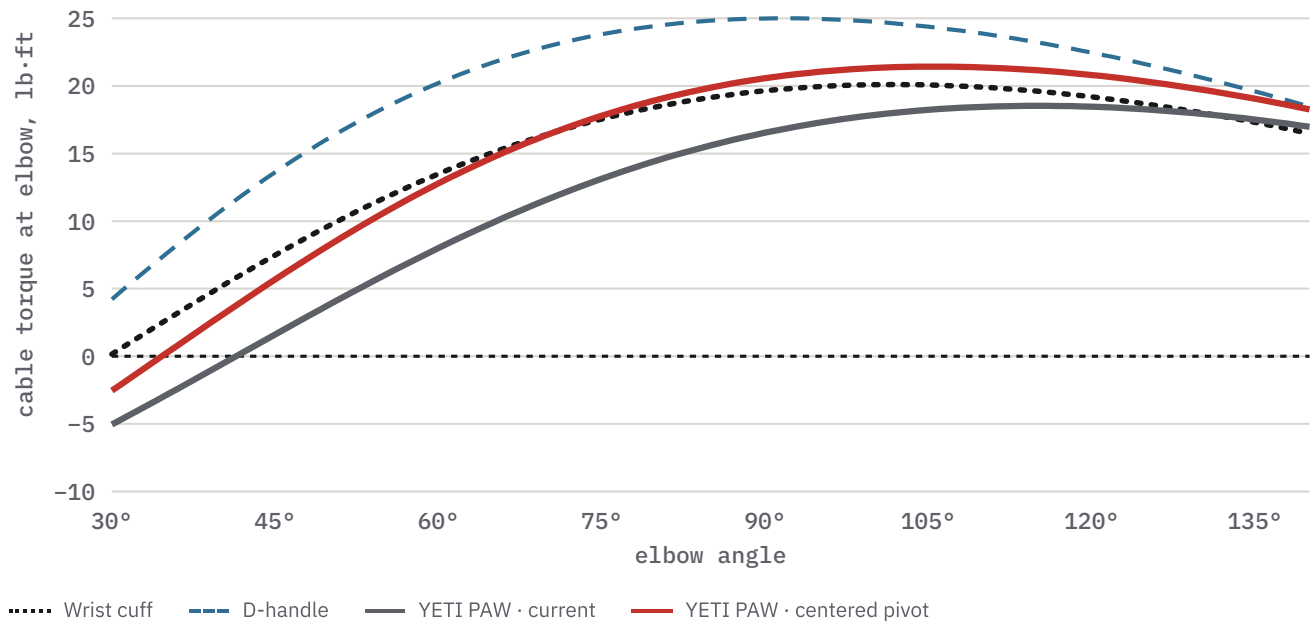
Reading the bars: red is the forearm pad's share of the cable, blue the hand's; each bar adds up to the 20 lb cable. A wrist cuff puts everything on the wrist strap.



Reading the chart: the same split through the whole movement. At every angle the solid line (forearm pad) and the dashed line of the same color (hand) add up to 20 lb.

AGAINST A CUFF AND A D-HANDLE

Two different comparisons, kept apart. **Where the load sits:** the current pivot, with most of it on the forearm pad, is nearer a wrist cuff; the centered pivot, sharing it with the grip, is nearer a D-handle. **The shape of the resistance curve:** see below. Where the cable hooks on shapes how the resistance builds and fades. With the cuff ring where we assumed it (3 in. behind the fist center, 1.2 in. below the wrist), the centered pivot's curve is the closer one: an RMS gap of 7.2% of the cuff's peak (centered) against 19.5% (current). Move the ring anywhere from 2.5–5 in. back and 0.8–2 in. down and the ranking flips: centered 2.1–22.3%, current 9.1–23.5%. So neither pivot can be said to follow a cuff more closely in general.



Reading the chart: cable torque at the elbow: how hard the cable turns the limb about the joint, in lb·ft (pounds times feet of leverage). The limb's own weight is not included. Below zero, the cable is pulling the limb along rather than resisting it.

At the start of the rep (30°), the stretched position, the four handles differ: cable torque at the elbow is −5.0 lb·ft with the current pivot, −2.5 centered, 0.2 with a wrist cuff and 4.2 with a D-handle. How much the stretched position is loaded is a real training difference between the handles, not a flaw in any of them.

Both to trial

Trial hypothesis, not a verdict. Both pivots go to the lifter trial as a hypothesis: the current pivot for lifters who want the forearm to carry the load, the centered pivot for a balanced, D-handle-like feel. A variant drops out if lifters can’t tell it apart from the other, if most prefer the other in a back-to-back trial, or if its pad lifts off and clunks where the numbers say it might.

APPENDIX · CABLE TORQUE NUMBERS

ANGLE	DUMBBELL	D-HANDLE	WRIST CUFF	CENTERED	CURRENT
30°	12	4	0	−3	−5
45°	18	14	7	6	2
60°	22	20	13	13	8
75°	24	24	18	18	13
90°	25	25	20	21	17
105°	24	24	20	21	18
120°	22	22	19	21	18
135°	18	20	17	19	18

Reading the table: cable torque at the elbow in lb·ft for a 20 lb load at each angle (limb weight not included). Negative means the cable is helping the movement at that angle. The dumbbell column is a 20 lb dumbbell, for comparison.

APPENDIX · HAND AND FOREARM, IN DETAIL

ANGLE	CURRENT · SHARE PAD / HAND	PAD PUSH	GRIP	CENTERED · SHARE PAD / HAND	PAD PUSH	GRIP
30°	0 / 20	15	24	1 / 19	11	22
45°	8 / 12	20	22	6 / 14	13	19
60°	16 / 4	23	17	10 / 10	14	14
75°	20 / −0.4	23	11	12 / 8	13	9
90°	20 / −0.4	21	3	11 / 9	11	9
105°	17 / 3	17	4	8 / 12	8	12
120°	11 / 9	12	10	4 / 16	5	16
135°	5 / 15	7	15	1 / 19	2	19

Reading the table: pounds. Share = each contact's part of the 20 lb cable, along the cable (the two add up to 20). Pad push and grip = the full contact forces, squeeze included. The squeeze is the pinch from the cable trying to twist the handle: you feel it, but it doesn’t add to the cable load. A negative pad push means the pad lifts off.

HANDLE	PEAK PAD PUSH, LB	PEAK GRIP FORCE, LB	PAD LIFTS OFF	WRIST MOMENT AT HARDEST, LB·IN	PEAK WRIST MOMENT, LB·IN
YETI PAW · current	23 at 67°	24 at 30°	never	17	42 at 30°
YETI PAW · centered	14 at 58°	22 at 30°	never	38	42 at 140°
D-handle	—	20 all rep	—	56	60 at 92°
Wrist cuff	—	0 strap 20	—	0	0 at 30°

Reading the table: 20 lb on the cable, over the whole rep. Pad push and grip force are the full contact forces, squeeze included, so they can add up to more than 20 lb (the squeeze cancels between them). The pad lifts off where its push turns negative. Wrist moment: the bending load the hand puts on the wrist joint, taken 3 in. behind the fist center along the forearm, in the plane of the movement (lb·in: pounds times inches of leverage). The cuff strap is taken to sit on the forearm side of the wrist, so it puts none on the wrist.

APPENDIX · HOW THESE NUMBERS WERE MADE

MEASURED

- Lifter 6′5″
- Shoulder joint 62″ off the floor, fist center 29″ with the arm hanging (33″ shoulder to fist)
- Handle holes: current 120 / 140 / 85 mm, centered 120 / 85 / 85 mm

ASSUMED

- Elbow to fist center 15″
- Voltra on a 45° rack bracket, cable exit 8″ up, 16″ in front of the elbow; elbow pinned at the side, 44″ up
- Movement in one plane (side view, facing the Voltra)
- Handle–pad line along the forearm; pad presses square to it on the side away from the cable; the grip does the rest
- Wrist cuff ring 3″ behind the fist center, 1.2″ below the wrist line; wrist joint 3″ behind the fist center

CHECKS

- Same statics as the lateral raise study (re-derived independently there; matched within 1%)
- Cable torque = cable force × its distance from the elbow; the cable pulls from the yoke pivot toward the cable exit
- Sensitivity: forearm length (13.5–16.5″) and stance (12–20″ from the elbow) moved; forearm share at the hardest point stays 12–14 lb (current) vs 5–6 lb (centered)
- Handle tilted ±10° and ±20° on the limb: 4–21 lb (current) vs 0–11 lb (centered)
- Cuff ring moved 2.5–5″ back, 0.8–2″ down: which pivot is closer to the cuff curve flips

Designed by Schnibbz · YETI STRENGTH · YETI PAW research v2.1 · 2026-10-17

YETI PAW · CABLE CURL · FACING AWAY FROM THE VOLTRA · DESIGN STUDY

BAYESIAN CURL: TWO WAYS TO CARRY THE LOAD

The same comparison as the lateral raise, on the Bayesian curl. The current pivot puts most of the load on the forearm pad. The centered pivot hooks on near the wrist and shares the load between hand and pad, closer to a D-ring handle. Both go to a lifter trial as a hypothesis.

THE BOTTOM LINE

13 + 7 lb

Current: forearm-led. At the hardest point of the Bayesian curl (89°) the forearm pad holds 13 lb of a 20 lb cable and the hand 7 lb. Over the rep the pad pushes up to 23 lb and the grip up to 24 lb.

6 + 14 lb

Centered: balanced. The same moment: 6 lb on the pad, 14 lb in the hand. Over the rep the pad pushes up to 14 lb and the grip up to 22 lb. A plain D-handle puts all 20 in the hand.

−14% weight

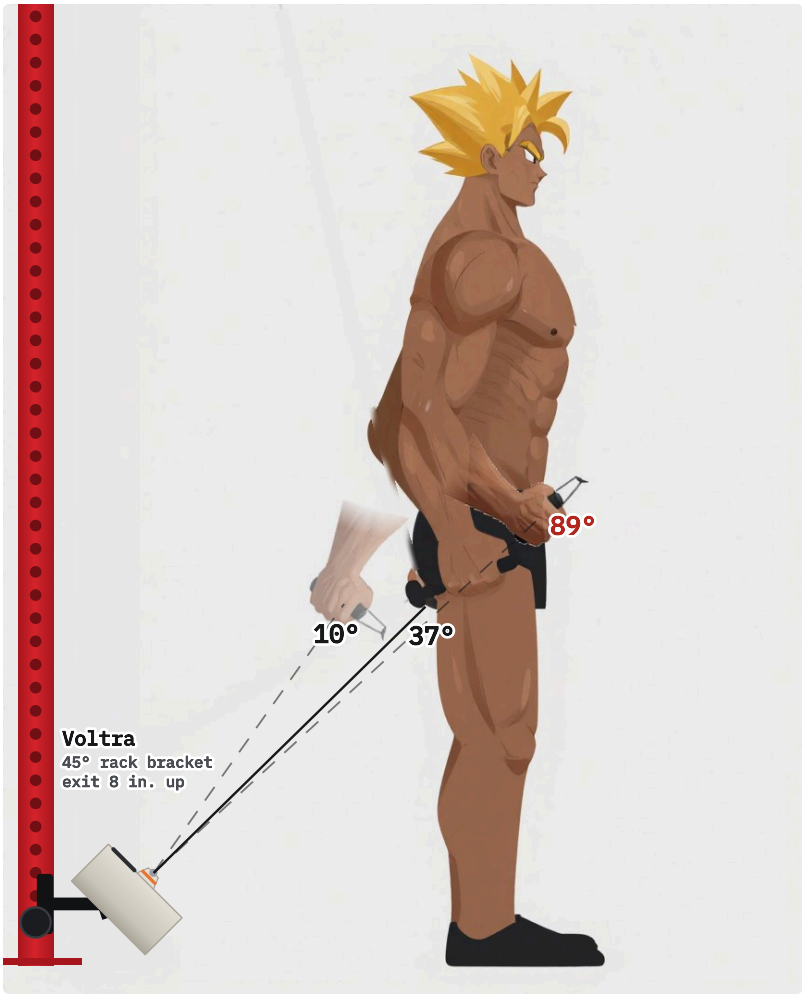
The same peak at about 14% less weight. At equal weight the centered pivot's peak cable torque at the elbow is 16% higher than the current pivot's (a wrist cuff: +8%), so the same peak takes about 14% less on the Voltra.

How sure are we? Confident on the direction and rough size, not on exact pounds or feel. The numbers come from the same 2D physics model as the lateral raise study, built on one real lifter (6′5″, measured), with the stance and forearm length assumed (see the appendix). Across the stances checked, the forearm holds 12–14 lb at the hardest point with the current pivot and 5–6 lb with the centered one. Tilting the handle 10–20° on the arm moves it more: 4–20 lb (current), 1–11 lb (centered).

The grip can relax mid-rep: at the hardest point it takes 8 lb (full grip force). Not at the ends: from 0–25°, 128–130° the current pivot's grip takes up to 24 lb, more than the 20 lb a D-handle puts in the hand. The current pivot's pad lifts off at 128–130° and the centered pivot's pad lifts off at 118–130°: the handle then hangs on the grip alone, and the pad lands again on the way back, which may be felt as a clunk. Wear the handle so the pad sits on the side of the forearm away from the cable, as on the lateral raise.

THE MOVEMENT

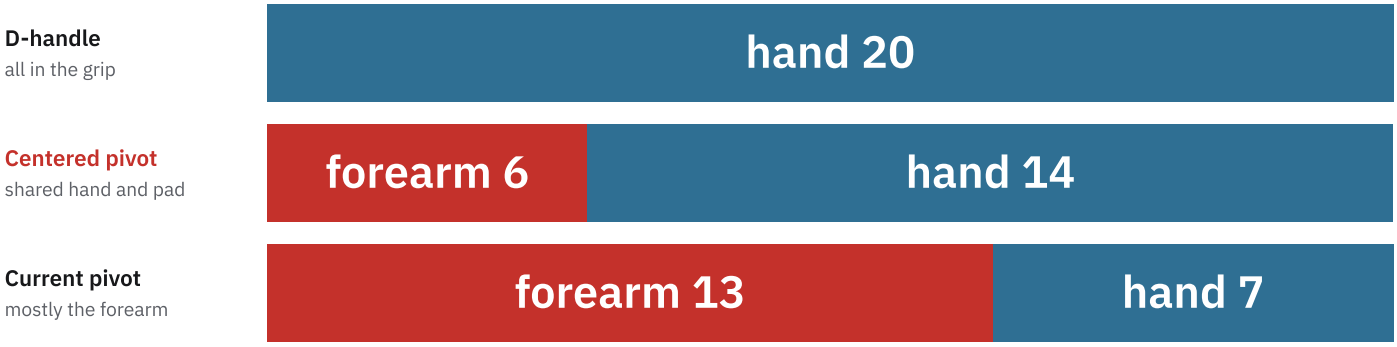
A Bayesian cable curl, seen from the side: facing away from the Voltra, with the upper arm held about 25° behind the body. The cable comes from a Voltra on a 45° rack bracket behind him: it exits 8 in. off the floor, about 28 in. behind the shoulder. The forearm curls from straight (0°) to 130°.



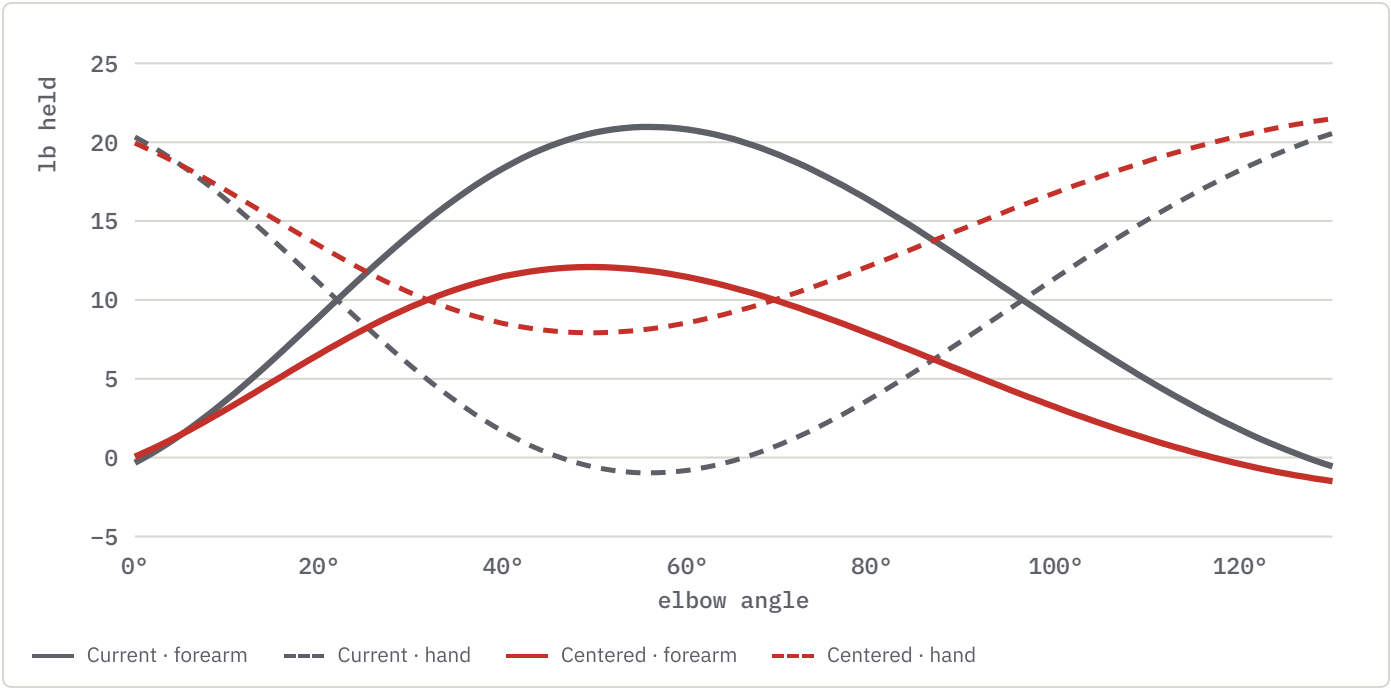
Drawn to the model: one forearm, one length, pivoting about one elbow at the exact angles shown (solid = the pose as drawn, faded = other points in the curl), and the Voltra's cable exit where the model puts it, 8 in. off the floor he stands on. The red angle is the hardest point.

WHERE THE 20 LB SITS

At the hardest point (89°), here is how a 20 lb cable is carried, with a plain D-handle for reference.



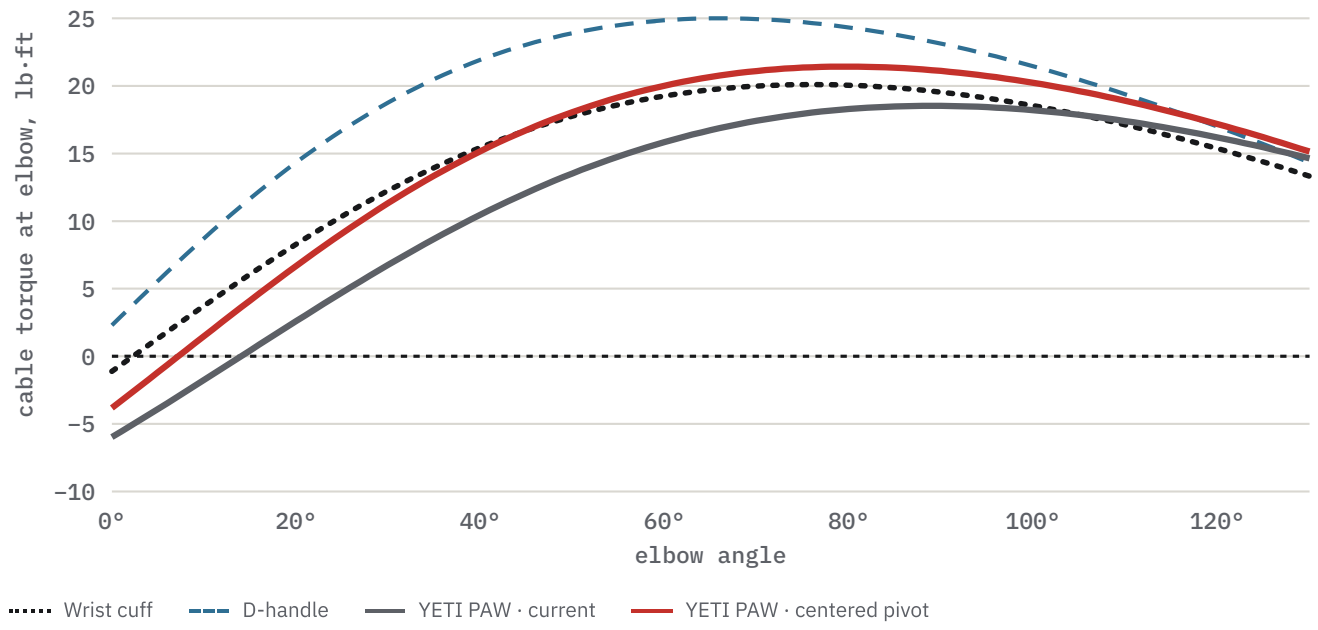
Reading the bars: red is the forearm pad's share of the cable, blue the hand's; each bar adds up to the 20 lb cable. A wrist cuff puts everything on the wrist strap.



Reading the chart: the same split through the whole movement. At every angle the solid line (forearm pad) and the dashed line of the same color (hand) add up to 20 lb.

AGAINST A CUFF AND A D-HANDLE

Two different comparisons, kept apart. **Where the load sits:** the current pivot, with most of it on the forearm pad, is nearer a wrist cuff; the centered pivot, sharing it with the grip, is nearer a D-handle. **The shape of the resistance curve:** see below. Where the cable hooks on shapes how the resistance builds and fades. With the cuff ring where we assumed it (3 in. behind the fist center, 1.2 in. below the wrist), the centered pivot's curve is the closer one: an RMS gap of 7.5% of the cuff's peak (centered) against 18.1% (current). Move the ring anywhere from 2.5–5 in. back and 0.8–2 in. down and the ranking flips: centered 2.2–22.2%, current 9.1–21.8%. So neither pivot can be said to follow a cuff more closely in general.



Reading the chart: cable torque at the elbow: how hard the cable turns the limb about the joint, in lb·ft (pounds times feet of leverage). The limb's own weight is not included. Below zero, the cable is pulling the limb along rather than resisting it.

At the start of the rep (0°), the stretched position, the four handles differ: cable torque at the elbow is −6.0 lb·ft with the current pivot, −3.8 centered, −1.1 with a wrist cuff and 2.3 with a D-handle. How much the stretched position is loaded is a real training difference between the handles, not a flaw in any of them.

Both to trial

Trial hypothesis, not a verdict. Both pivots go to the lifter trial as a hypothesis: the current pivot for lifters who want the forearm to carry the load, the centered pivot for a balanced, D-handle-like feel. A variant drops out if lifters can’t tell it apart from the other, if most prefer the other in a back-to-back trial, or if its pad lifts off and clunks where the numbers say it might.

APPENDIX · CABLE TORQUE NUMBERS

ANGLE	D-HANDLE	WRIST CUFF	CENTERED	CURRENT
0°	2	−1	−4	−6
15°	12	6	4	0
30°	19	12	11	7
45°	23	17	17	12
60°	25	19	20	16
75°	25	20	21	18
90°	23	20	21	19
105°	20	18	20	18
120°	17	15	17	16

Reading the table: cable torque at the elbow in lb·ft for a 20 lb load at each angle (limb weight not included). Negative means the cable is helping the movement at that angle.

APPENDIX · HAND AND FOREARM, IN DETAIL

ANGLE	CURRENT · SHARE PAD / HAND	PAD PUSH	GRIP	CENTERED · SHARE PAD / HAND	PAD PUSH	GRIP
0°	−0.3 / 20	14	24	0 / 20	10	22
15°	6 / 14	19	23	5 / 15	13	19
30°	14 / 6	23	19	10 / 10	14	15
45°	20 / 0	23	12	12 / 8	14	10
60°	21 / −1	21	5	11 / 9	12	9
75°	18 / 2	18	2	9 / 11	9	11
90°	13 / 7	13	9	5 / 15	6	15
105°	7 / 13	8	14	2 / 18	3	18
120°	2 / 18	3	18	−0.4 / 20	−1 lifts off	20

Reading the table: pounds. Share = each contact's part of the 20 lb cable, along the cable (the two add up to 20). Pad push and grip = the full contact forces, squeeze included. The squeeze is the pinch from the cable trying to twist the handle: you feel it, but it doesn't add to the cable load. A negative pad push means the pad lifts off.

HANDLE	PEAK PAD PUSH, LB	PEAK GRIP FORCE, LB	PAD LIFTS OFF	WRIST MOMENT AT HARDEST, LB·IN	PEAK WRIST MOMENT, LB·IN
YETI PAW · current	23 at 41°	24 at 0°	128–130°	17	42 at 0°
YETI PAW · centered	14 at 32°	22 at 0°	118–130°	38	46 at 130°
D-handle	—	20 all rep	—	56	60 at 66°
Wrist cuff	—	0 strap 20	—	0	0 at 0°

Reading the table: 20 lb on the cable, over the whole rep. Pad push and grip force are the full contact forces, squeeze included, so they can add up to more than 20 lb (the squeeze cancels between them). The pad lifts off where its push turns negative. Wrist moment: the bending load the hand puts on the wrist joint, taken 3 in. behind the fist center along the forearm, in the plane of the movement (lb·in: pounds times inches of leverage). The cuff strap is taken to sit on the forearm side of the wrist, so it puts none on the wrist.

APPENDIX · HOW THESE NUMBERS WERE MADE

MEASURED

- Lifter 6′5″
- Shoulder joint 62″ off the floor, fist center 29″ with the arm hanging (33″ shoulder to fist)
- Handle holes: current 120 / 140 / 85 mm, centered 120 / 85 / 85 mm

ASSUMED

- Elbow to fist center 15″
- Voltra on a 45° rack bracket, cable exit 8″ up, 28″ behind the shoulder; upper arm held 25° behind vertical
- Movement in one plane (side view, facing away from the Voltra)
- Handle–pad line along the forearm; pad presses square to it on the side away from the cable; the grip does the rest
- Wrist cuff ring 3″ behind the fist center, 1.2″ below the wrist line; wrist joint 3″ behind the fist center

CHECKS

- Same statics as the lateral raise study (re-derived independently there; matched within 1%)
- Cable torque = cable force × its distance from the elbow; the cable pulls from the yoke pivot toward the cable exit
- Sensitivity: forearm length (13.5–16.5″), upper-arm angle (15–35° back) and stance (20–36″ behind) moved; forearm share at the hardest point stays 12–14 lb (current) vs 5–6 lb (centered)
- Handle tilted ±10° and ±20° on the limb: 4–20 lb (current) vs 1–11 lb (centered)
- Cuff ring moved 2.5–5″ back, 0.8–2″ down: which pivot is closer to the cuff curve flips

Designed by Schnibbz · YETI STRENGTH · YETI PAW research v2.1 · 2026-10-17

YETI PAW · CABLE FLY · PAIR OF CABLES · DESIGN STUDY

CABLE CHEST FLY: TWO WAYS TO CARRY THE LOAD

The same comparison as the lateral raise, on the cable chest fly. The current pivot puts most of the load on the forearm pad. The centered pivot hooks on near the wrist and shares the load between hand and pad, closer to a D-ring handle. Both go to a lifter trial as a hypothesis.

THE BOTTOM LINE

17 + 3 lb

Current: forearm-led. At the hardest point of the cable chest fly (31°) the forearm pad holds 17 lb of a 20 lb cable and the hand 3 lb. Over the rep the pad pushes up to 23 lb and the grip up to 24 lb.

8 + 12 lb

Centered: balanced. The same moment: 8 lb on the pad, 12 lb in the hand. Over the rep the pad pushes up to 14 lb and the grip up to 21 lb. A plain D-handle puts all 20 in the hand.

−6% weight

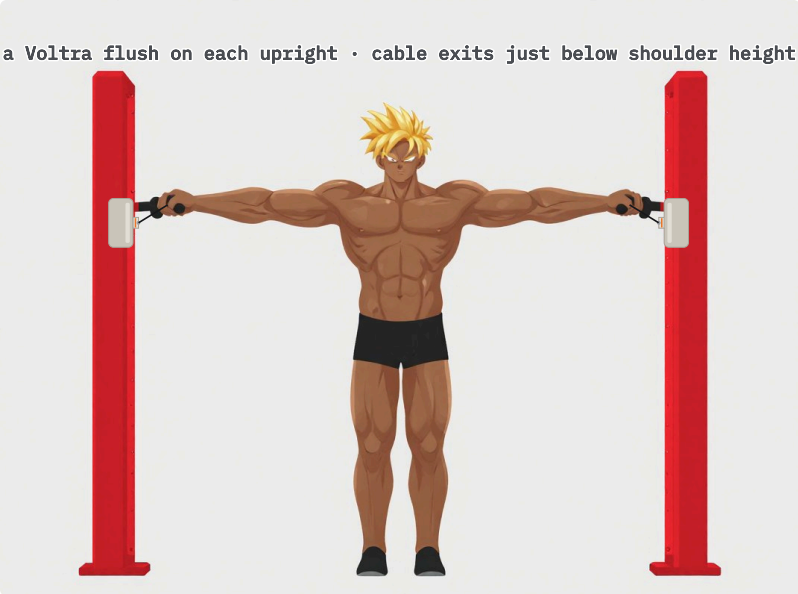
The same peak at about 6% less weight. At equal weight the centered pivot's peak cable torque at the shoulder is 7% higher than the current pivot's (a wrist cuff: +4%), so the same peak takes about 6% less on the Voltra.

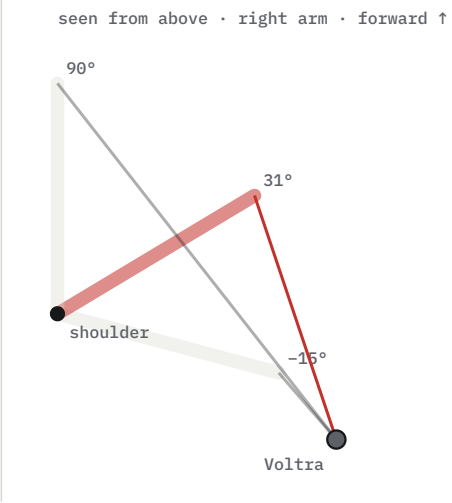
How sure are we? Confident on the direction and rough size, not on exact pounds or feel. The numbers come from the same 2D physics model as the lateral raise study, built on one real lifter (6′5″, measured), with the cable-exit position assumed (see the appendix). Across the stances checked, the forearm holds 17 lb at the hardest point with the current pivot and 8 lb with the centered one. Tilting the handle 10–20° on the arm moves it more: 8–21 lb (current), 3–12 lb (centered). The arm is modeled straight; a fly is usually done with a soft elbow, which is not modeled.

The grip can relax mid-rep: at the hardest point it takes 4 lb (full grip force). Not at the ends: from −15° to −9° the current pivot's grip takes up to 24 lb, more than the 20 lb a D-handle puts in the hand. The centered pivot's pad lifts off at 79–90°: the handle then hangs on the grip alone, and the pad lands again on the way back, which may be felt as a clunk. Wear the handle so the pad sits on the side of the forearm away from the cable, as on the lateral raise.

THE MOVEMENT

A standing cable chest fly between a pair of Voltras flush-mounted on uprights, cable exits just below shoulder height, about 4 ft out to each side and 18 in. behind the shoulders. Each arm works the same way, so the sheet follows one arm, seen from above, as it sweeps from slightly behind the body (−15°) to pointing straight ahead (90°).

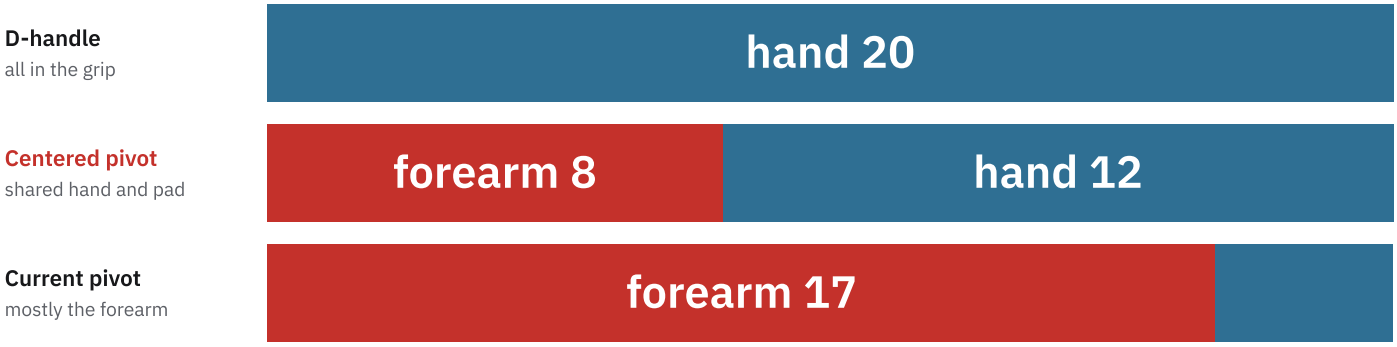




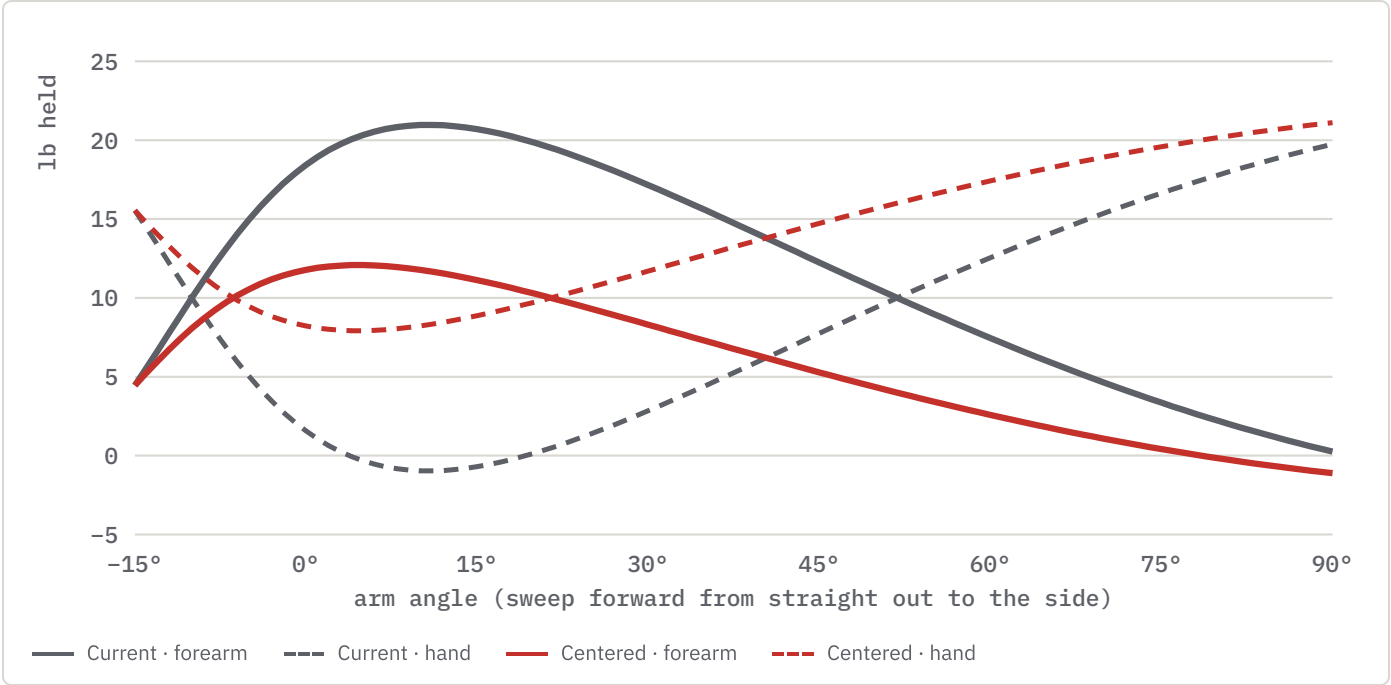
From the front: both cables come from Voltras flush-mounted on the uprights, just below shoulder height. The sweep happens in depth, so the inset shows it from above, drawn from the model: −15° (stretched, start), the hardest point, and 90° (arms straight ahead).

WHERE THE 20 LB SITS

At the hardest point (31°), here is how a 20 lb cable is carried, with a plain D-handle for reference.



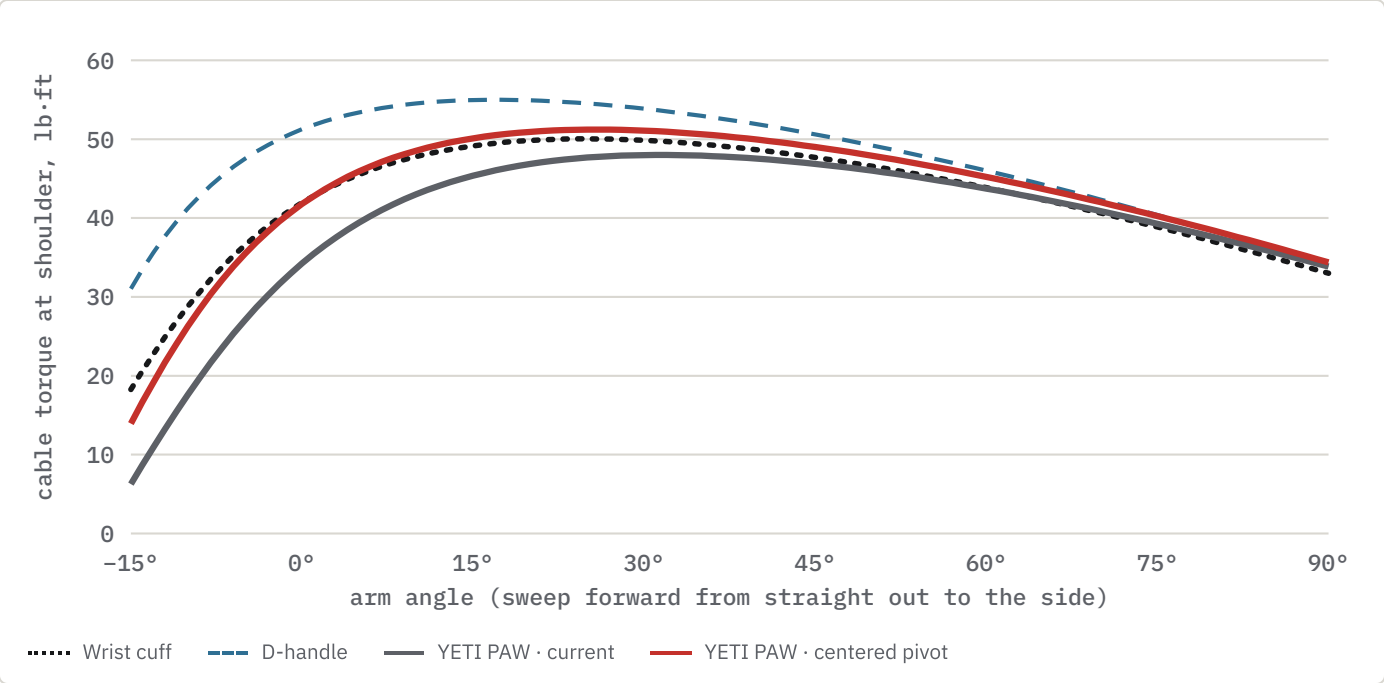
Reading the bars: red is the forearm pad's share of the cable, blue the hand's; each bar adds up to the 20 lb cable. A wrist cuff puts everything on the wrist strap.



Reading the chart: the same split through the whole movement. At every angle the solid line (forearm pad) and the dashed line of the same color (hand) add up to 20 lb.

AGAINST A CUFF AND A D-HANDLE

Two different comparisons, kept apart. **Where the load sits:** the current pivot, with most of it on the forearm pad, is nearer a wrist cuff; the centered pivot, sharing it with the grip, is nearer a D-handle. **The shape of the resistance curve:** see below. Where the cable hooks on shapes how the resistance builds and fades. With the cuff ring where we assumed it (3 in. behind the fist center, 1.2 in. below the wrist), the centered pivot's curve is the closer one: an RMS gap of 2.9% of the cuff's peak (centered) against 9.3% (current). Move the ring anywhere from 2.5–5 in. back and 0.8–2 in. down and the ranking flips: centered 0.9–9.1%, current 3.7–11.4%. So neither pivot can be said to follow a cuff more closely in general.



Reading the chart: cable torque at the shoulder: how hard the cable turns the limb about the joint, in lb·ft (pounds times feet of leverage). The limb's own weight is not included.

At the start of the rep (−15°), the stretched position, the four handles differ: cable torque at the shoulder is 6.3 lb·ft with the current pivot, 13.9 centered, 18.3 with a wrist cuff and 31.0 with a D-handle. How much the stretched position is loaded is a real training difference between the handles, not a flaw in any of them.

Both to trial

Trial hypothesis, not a verdict. Both pivots go to the lifter trial as a hypothesis: the current pivot for lifters who want the forearm to carry the load, the centered pivot for a balanced, D-handle-like feel. A variant drops out if lifters can’t tell it apart from the other, if most prefer the other in a back-to-back trial, or if its pad lifts off and clunks where the numbers say it might.

APPENDIX · CABLE TORQUE NUMBERS

ANGLE	D-HANDLE	WRIST CUFF	CENTERED	CURRENT
−15°	31	18	14	6
0°	51	42	42	34
15°	55	49	50	45
30°	54	50	51	48
45°	51	48	49	47
60°	46	44	45	44
75°	40	39	40	39
90°	34	33	34	34

Reading the table: cable torque at the shoulder in lb·ft for a 20 lb load at each angle (limb weight not included).

APPENDIX · HAND AND FOREARM, IN DETAIL

ANGLE	CURRENT · SHARE PAD / HAND	PAD PUSH	GRIP	CENTERED · SHARE PAD / HAND	PAD PUSH	GRIP
−15°	4 / 16	18	24	4 / 16	13	20
0°	18 / 2	23	14	12 / 8	14	11
15°	21 / −1	21	5	11 / 9	11	9
30°	17 / 3	17	3	8 / 12	8	12
45°	12 / 8	13	9	5 / 15	6	15
60°	7 / 13	9	13	3 / 17	3	17
75°	3 / 17	4	17	0 / 20	1	20
90°	0 / 20	0	20	−1 / 21	−2 lifts off	21

Reading the table: pounds. Share = each contact's part of the 20 lb cable, along the cable (the two add up to 20). Pad push and grip = the full contact forces, squeeze included. The squeeze is the pinch from the cable trying to twist the handle: you feel it, but it doesn't add to the cable load. A negative pad push means the pad lifts off.

HANDLE	PEAK PAD PUSH, LB	PEAK GRIP FORCE, LB	PAD LIFTS OFF	WRIST MOMENT AT HARDEST, LB·IN	PEAK WRIST MOMENT, LB·IN
YETI PAW · current	23 at 0°	24 at −15°	never	9	40 at −15°
YETI PAW · centered	14 at −6°	21 at 90°	79–90°	35	45 at 90°
D-handle	—	20 all rep	—	59	60 at 17°
Wrist cuff	—	0 strap 20	—	0	0 at −15°

Reading the table: 20 lb on the cable, over the whole rep. Pad push and grip force are the full contact forces, squeeze included, so they can add up to more than 20 lb (the squeeze cancels between them). The pad lifts off where its push turns negative. Wrist moment: the bending load the hand puts on the wrist joint, taken 3 in. behind the fist center along the forearm, in the plane of the movement (lb·in: pounds times inches of leverage). The cuff strap is taken to sit on the forearm side of the wrist, so it puts none on the wrist.

APPENDIX · HOW THESE NUMBERS WERE MADE

MEASURED

- Lifter 6′5″
- Shoulder joint 62″ off the floor, fist center 29″ with the arm hanging (33″ shoulder to fist)
- Handle holes: current 120 / 140 / 85 mm, centered 120 / 85 / 85 mm

ASSUMED

- Straight arm, 33″ shoulder to fist
- Voltras flush-mounted, cable exits just below shoulder height, 48″ out from the midline and 18″ behind the shoulders; straight arm
- Movement in one plane (seen from above, one arm of the pair); arm straight, no elbow bend
- Handle–pad line along the forearm; pad presses square to it on the side away from the cable; the grip does the rest
- Wrist cuff ring 3″ behind the fist center, 1.2″ below the wrist line; wrist joint 3″ behind the fist center

CHECKS

- Same statics as the lateral raise study (re-derived independently there; matched within 1%)
- Cable torque = cable force × its distance from the shoulder; the cable pulls from the yoke pivot toward the cable exit
- Sensitivity: cable-exit position (40–56″ out, 8–28″ behind the shoulders) and arm length (30″) moved; forearm share at the hardest point stays 17 lb (current) vs 8 lb (centered)
- Handle tilted ±10° and ±20° on the limb: 8–21 lb (current) vs 3–12 lb (centered)
- Cuff ring moved 2.5–5″ back, 0.8–2″ down: which pivot is closer to the cuff curve flips

Designed by Schnibbz · YETI STRENGTH · YETI PAW research v2.1 · 2026-10-17

YETI PAW · FROM STUDY TO HARDWARE

THE CENTERED PLATE, CHECKED ON EVERY MOVEMENT

The plate outline, the yoke's swing clearance and the carabiner eye, checked against the study.

V2.1 · 2026-10-17

SIDE PLATE

4.4

Static, both plates. The centered plate matches the original in the same 2D check (100 lb cable, ELEGOO PETG-CF).

1.5–1.6

Fatigue. Original 1.5, centered 1.6. Yoke hole 85 mm from handle and pad; yoke side swept R20.

75 lb

Rating. The yoke, not the plate, is the weakest part. A Voltra can be set above 75 lb, so don't.

YOKE SWING

The yoke follows the cable (plus the 5.4° eye offset); the plate lets it swing −81° … +103° from its rest line (+ toward the handle). "Needs" covers the baseline and every stance in that movement's study. Past the stop the yoke rides on it and the eye tilts further, with the cable still at full tension: a small cable torque at the joint early in the rep does not mean a small load on the handle.

MOVEMENT	YOKE NEEDS, ALL STANCES	PAST THE STOP
Lateral raise	−44° … +36°	no
Bicep curl	−42° … +99°	no
Bayesian curl	−62° … +115°	the upper arm 35° back: 0–8° of the rep, 12° over; the cable exit 20″ away: 0–5° of the rep, 7° over
Cable chest fly	−53° … +122°	the cable exit 8″ behind: −15° to −10° of the rep, 19° over

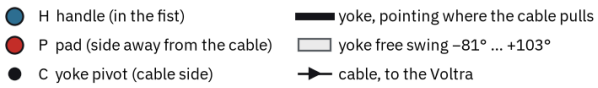
CARABINER EYE

One plain bronze sleeve, flush both faces. McMaster 6391K256, oil-embedded SAE 841, 3/4″ bore × 7/8″ OD × 1/2″ long, pressed through the 1/2″ eye, so both rims the carabiner bears on are bronze.

Contact model: bronze only up to 40° of tilt; the worst tilt in use is about 24° (the 5.4° eye offset plus up to 19° past the stop, at the start of some stances).

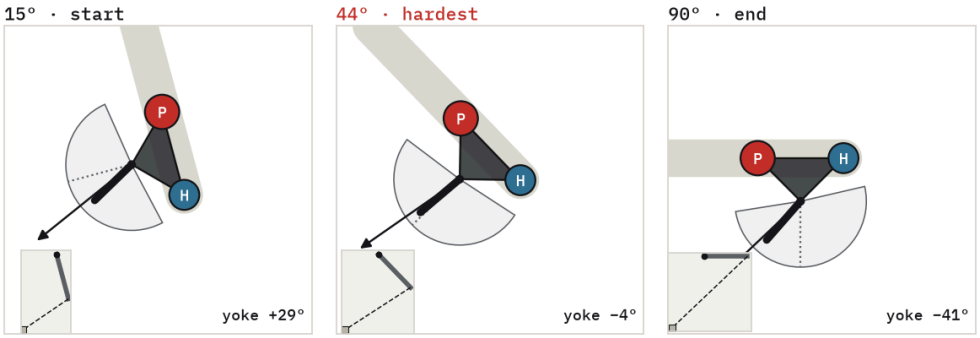
ORIENTATION CHECK

Each movement in its own view at the start, the hardest point and the end, drawn from the same model as the numbers (baseline stance): where the handle, pad and pivot sit, and where the cable points the yoke inside its free swing. The small inset shows the whole limb and the Voltra.



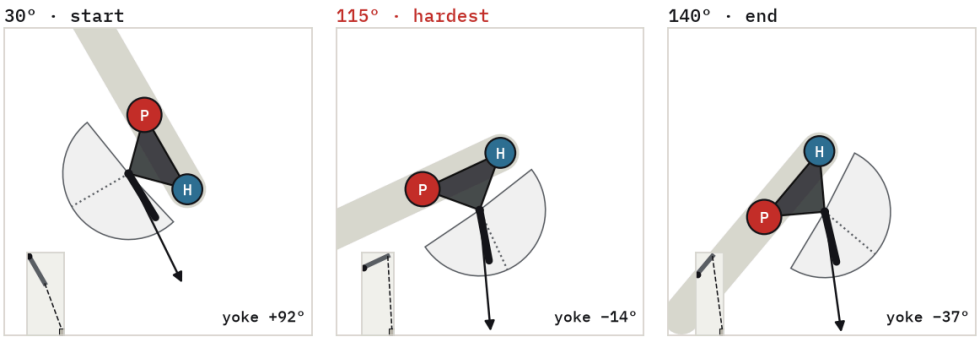
LATERAL RAISE

seen from behind

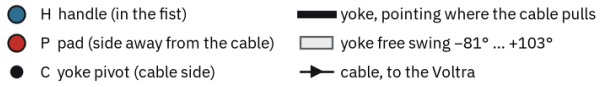


BICEP CURL

seen from his right side

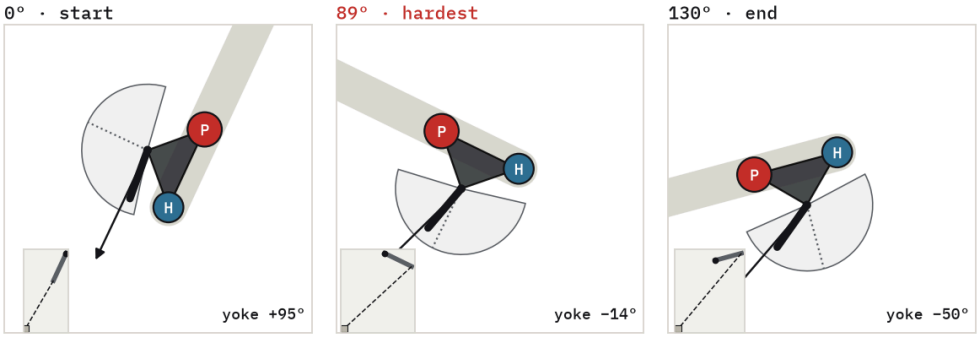


ORIENTATION CHECK, CONTINUED



BAYESIAN CURL

seen from his right side



CABLE CHEST FLY

seen from above

